

Tommaso Derossi

Data Scientist & Applied Researcher

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Summary

6+ years of experience in collaborating with the most diverse stakeholders, building **data-driven systems** to translate messy data into information-rich outputs. Currently focused on research projects to advance **manufacturing systems** methodologies through **classical and generative AI**.

Domains: Research | Government | NGOs | Enterprises | Startups

Stack: Docker | AWS (S3/EC2/Lambda) | Spark | Python [Pytorch - Fastapi - Flask - MLflow - Metaflow - W&B - Airflow] | R | Java | Javascript

Experience

GEA Group - GEA Procomac S.P.A.

November 2024 - Present

Industrial PhD

https://www.gea.com/en/

- Developing a data-driven recommendation system for **industrial troubleshooting** support.

Vector databases, Bayesian statistics, clustering algorithms, agentic patterns, local LLMs.

Supervised Program for Alignment Research

September 2025 - March 2026

Research Fellow

Remote

https://sparai.org/

- AI Alignment** research project on **Chain of Thought (CoT) monitoring** (supervised by Rauno Arike): development of **CoT decomposition** and **attribution** methods to study unsafe/strange LLM reasoning traces.

Freelance

September 2024 - February 2026

AI Engineer

- Developed a **recommendation/ranking system** for job-candidate matching.

LLM evaluations, CI/CD practices

Tomorrow S.R.L.

February 2023 - July 2024

Data Scientist

https://www.tomorrowssl.com/

- Conceptualized and developed data transformation pipelines integrated with **machine learning** models (Bayesian hierarchical models, Random forest, Gradient boosting) for the analysis and presentation of **IOT data** (Predictive maintenance, KPIs optimization, Data-driven actionable insights, Time series forecasting)
- Designed and implemented customized **AI conversational agent** for purchasing assistance based on **LLMs** (LLMs prompting and finetuning, RAG, LLamaIndex, Langhchain)
- Prototyped a **recommendation system** and visualization dashboard for data-driven soccer scouting decision-making support, based on wide high frequency **match-events data** and **machine learning** models (Pytorch, Sklearn)
- Created **marketing automation pipelines** from **target audience selection** and **content production** to **campaign creation and management** with **human-in-the-loop** for different advertisement platforms (Linkedin, Meta, X)

WATERONMARS S.R.L.

March 2022 - February 2023

Data Scientist

https://www.wateronmars.it/

- Built **ETL pipelines** (Visokio Omniscopes) for the download (Web-scraping, API) and analysis of web content, in particular, **social media data** (textual data, images, videos, geospatial and temporal data)
- Implemented and integrated **predictive models** for data enrichment in the domains of **computer vision** (OpenCV, SimpleCV) and **natural language processing** (Spacy, Transformers, NLTK, Gensim, TextBlob, Prodigy)
- Created highly customized **data visualization dashboards** for various clients and types of analysis (Geospatial analytics - GIS, Election campaign analysis, Brand monitoring, Buyer persona, USP analysis, E-commerce analytics)

OMIGRADE S.R.L.

March 2019 - October 2020

Machine Learning engineer

https://www.omigrate.it/

- Developed a **Recurrent Neural Network (LSTM model)** for an interactive classification of DB columns web application; engineered distributed retraining/monitoring pipeline/batch inference on Hadoop with Spark RDDs and **PySpark** processing.

MLOps: standardized data prep, periodic retraining, monitoring, inference, performance analysis.

Education

Università degli studi di Parma

November 2024 - Present

Artificial Intelligence

PhD Student

- Research interests in: **AI applications in industrial troubleshooting systems**, **Time series analysis**, **Large Language Models safety and interpretability**.

Università degli studi di Torino

September 2019 - April 2023

Stochastics and Data Science

Master Degree

https://www.unito.it/

- Degree course in **Mathematics** fully taught in **English**

Thesis: "Modelling the discourse around nature in Twitter in presence of highly imbalanced data."

Thesis topics: Natural Language Processing, Transformers models, Network Science, Filtering Techniques, Sentiment Classification, Semantic Clustering, Topic Modeling

KU Leuven

September 2020 - July 2021

Statistics

1 year exchange programme

https://www.kuleuven.be/

- Second year of master’s degree spent in Belgium at KU Leuven. Some of the main courses attended (10 courses in total): **Optimization and Numerical Methods** (17/20), **Concepts of Bayesian Data Analysis** (16/20), **Data Mining and Neural Networks** (15/20), and **Modern Data Analytics** (15/20)
- Participated in **Data4Good** Challenge (**Emergent Leuven**)

Università degli studi di Parma

September 2016 - September 2019

Physics

Bachelor Degree

https://www.unipr.it/

- Awarded** for being the **top-ranked student in credits earned relative to grade point average** during the second academic year

Thesis: "Methodological aspects of neural networks integrated within an application for the classification of relational database columns."

Thesis topics: Artificial Neural Networks, Deep Recurrent Neural Networks (LSTM), Distributed Model Training

Liceo Scientifico "Giacomo Ulivi"

2011 - 2016

Scientific Oriented

High School

South African College High School

2012

English School Attendance, 3 months

Skills			
Python	R	MATLAB	SQL, NoSQL
MS Office (Excel), LaTeX	ML/DL (Pytorch, Tensorflow)	Spark, PySpark	Visokio Omniscopes
Docker, Git	AWS (S3, EC2, Lambda)	LLM Evaluation	ML Flow, Weights&Biases, Metaflow, Airflow

Languages		
Italian	English	Spanish
Mother tongue	Fluent	Intermediate